Exact POMDP Solutions: α-vectors

Recap

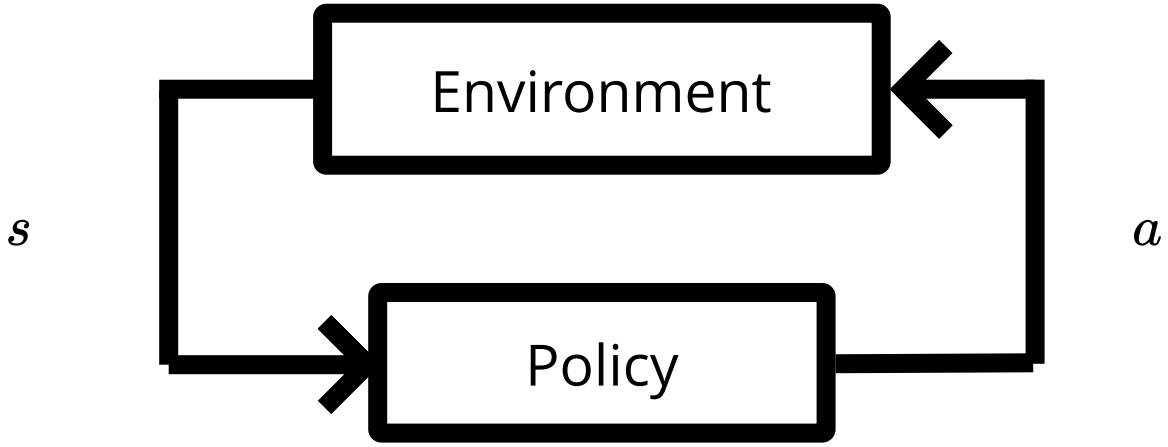
- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

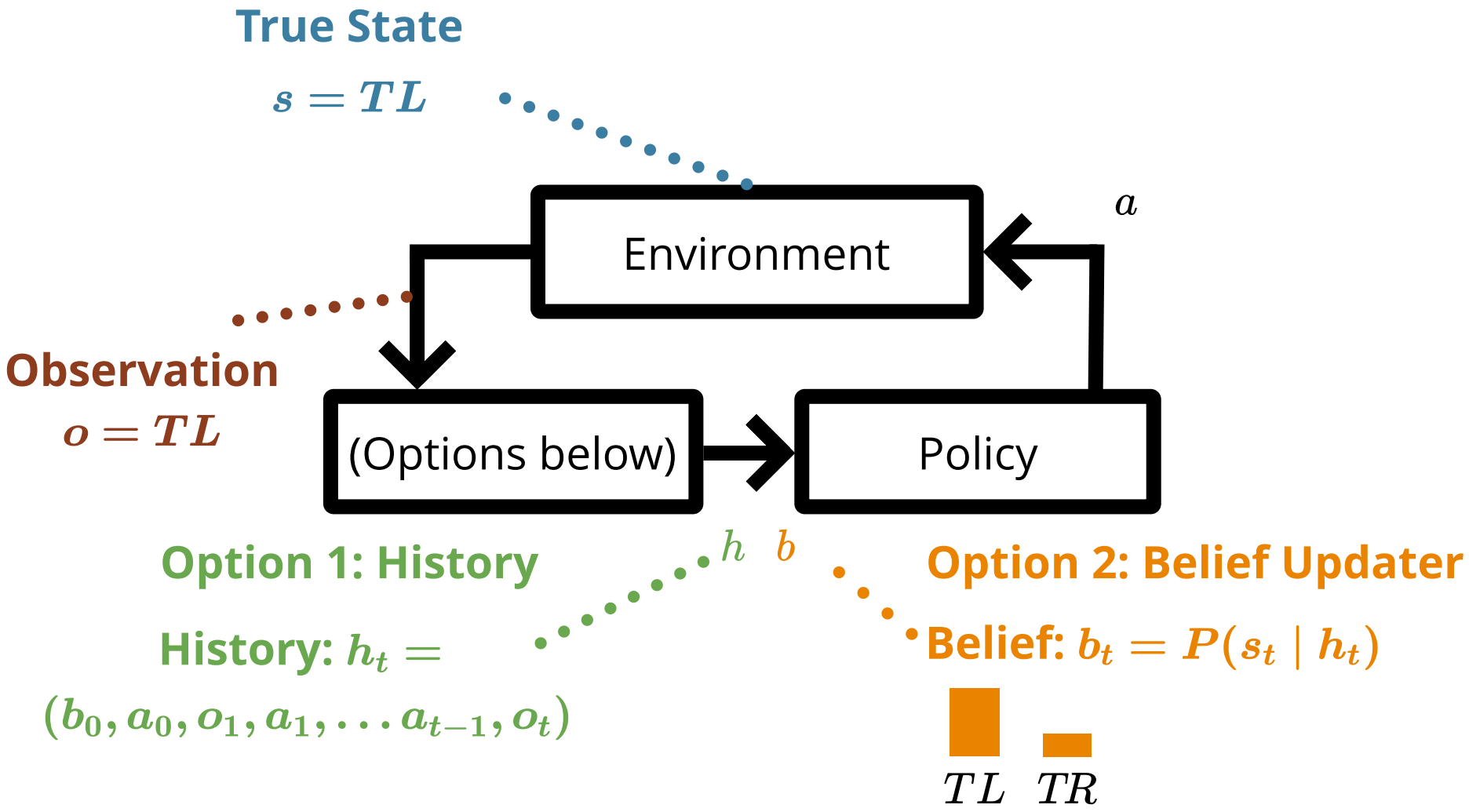
$$b' = \tau(b, a, o)$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a)\, b(s)$$

MDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



Exercise 1: Crying Baby Belief Update

$S = \{h, \neg h\}$

$A = \{f, \neg f\}$

$O = \{c, \neg c\}$

$T(h \mid h, \neg f) = 1.0$

$T(h \mid \neg h, \neg f) = 0.1$

$T(\neg h \mid \cdot, f) = 1.0$

$R(s, a) = R(s) + R(a)$

$$R(s) = \begin{cases} -10 \text{ if } s = h \\ 0 \text{ otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 \text{ if } a = f \\ 0 \text{ otherwise} \end{cases}$$

$Z(c \mid \cdot, h) = 0.8)$

$Z(c \mid \cdot, \neg h) = 0.1$

$\gamma = 0.9$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a)\, b(s)$$

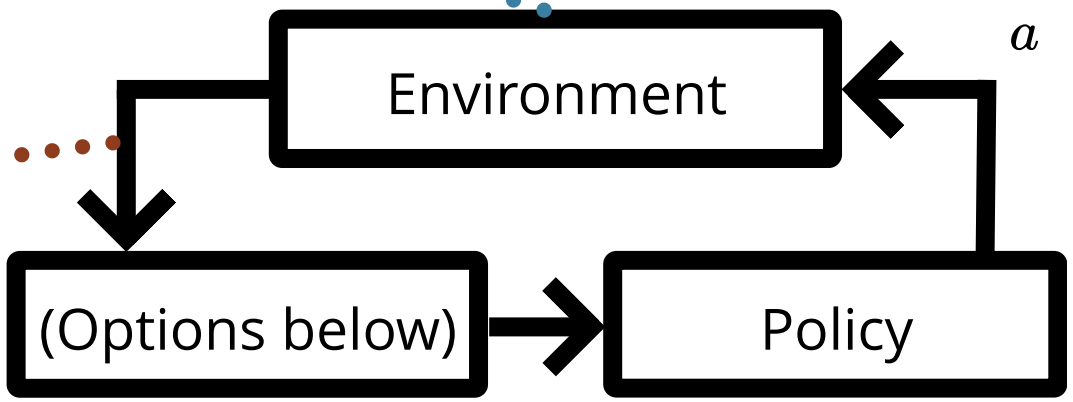
Starting at a $b(h) = 0$, calculate b' with $a = \neg f$ and $o = c$.

Guiding Quesiton

How do we calculate the optimal action in a POMDP?

True State

$s = TL$



Observation

$o = TL$

h b

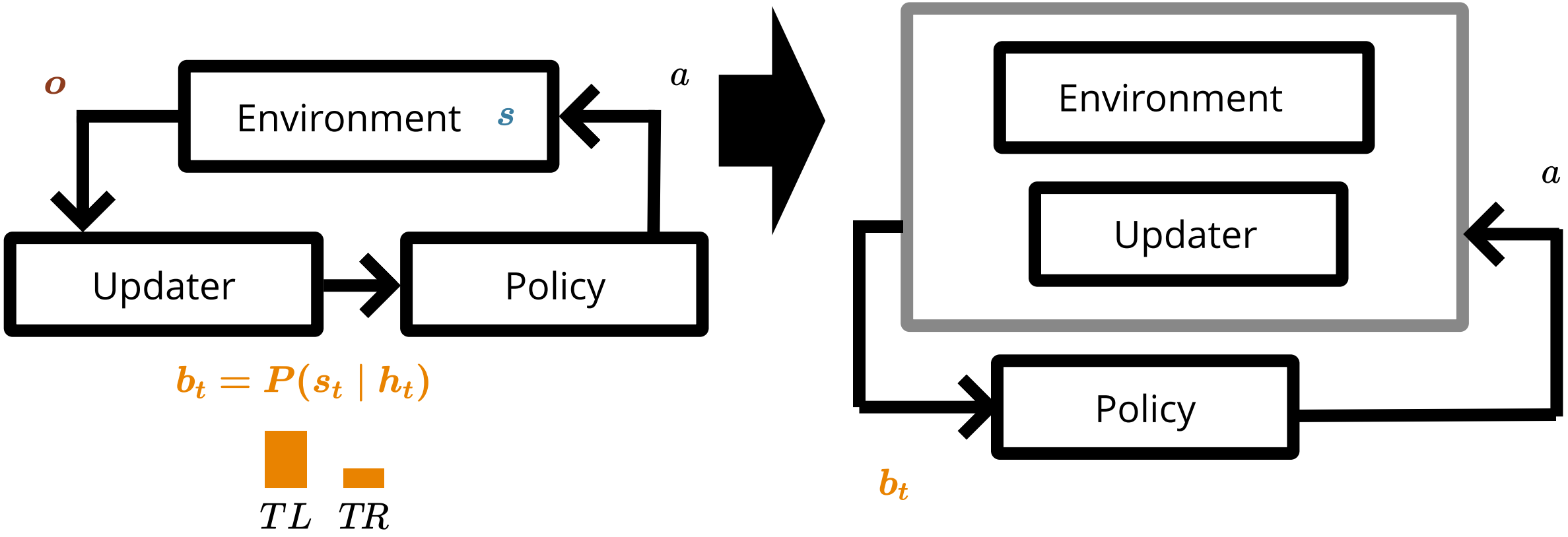
Option 1: History

History: $h_t = (b_0, a_0, o_1, a_1, \ldots a_{t-1}, o_t)$

Option 2: Belief Updater

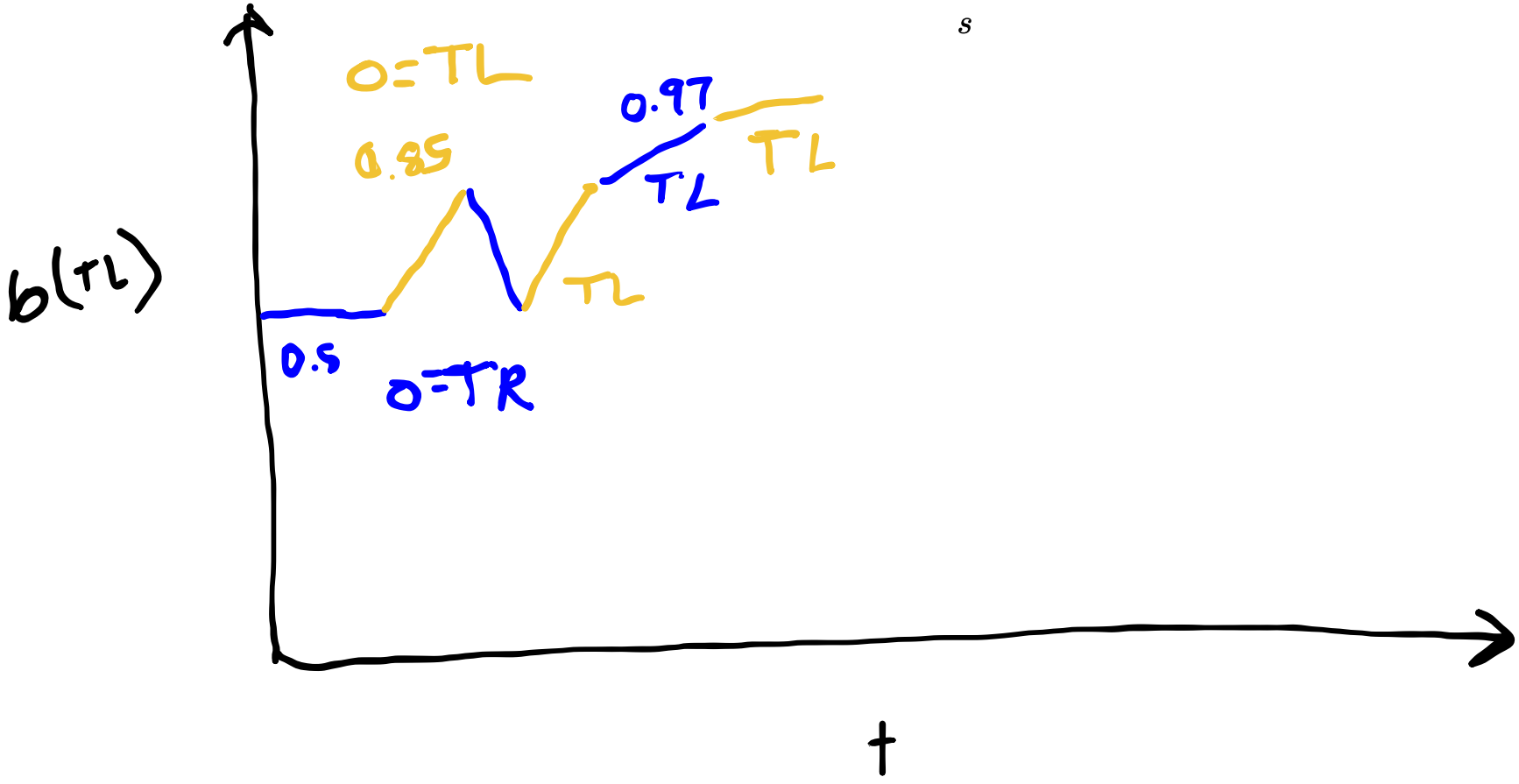
Belief: $b_t = P(s_t \mid h_t)$

Belief-Space MDP



Belief Dynamics

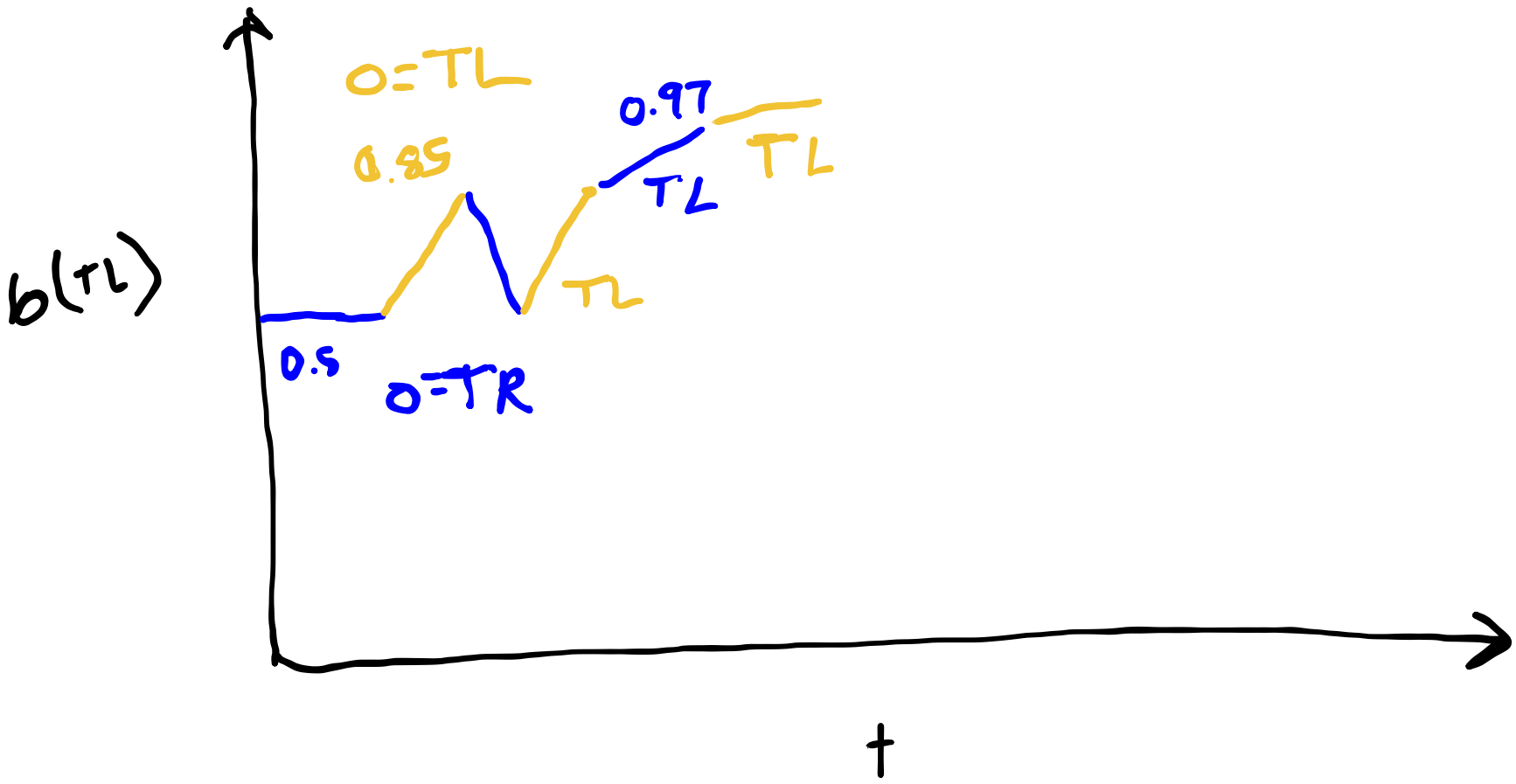
$$b'(s') \propto Z(o \mid a, s') \sum_{s} T(s' \mid s, a)\, b(s)$$



One-step utility

α-vectors define a policy

$$b'(s') \propto Z(o \mid a, s') \sum_{s} T(s' \mid s, a)\, b(s)$$



Exercise 2: Crying Baby 1-Step Utility

$S = \{h, \neg h\}$ $\quad T(h \mid h, \neg f) = 1.0$

$A = \{f, \neg f\}$ $\quad T(h \mid \neg h, \neg f) = 0.1$

$O = \{c, \neg c\}$ $\quad T(\neg h \mid \cdot, f) = 1.0$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 \text{ if } s = h \\ 0 \text{ otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 \text{ if } a = f \\ 0 \text{ otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8)$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

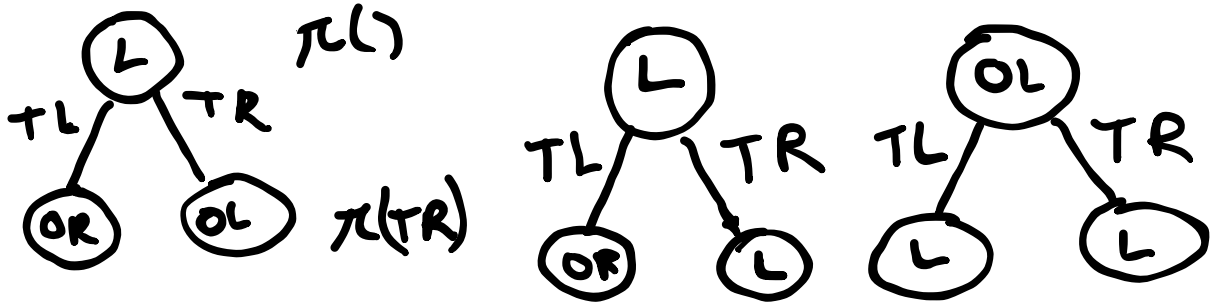
Draw the 1-step utility α-vectors for the Crying Baby problem.

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

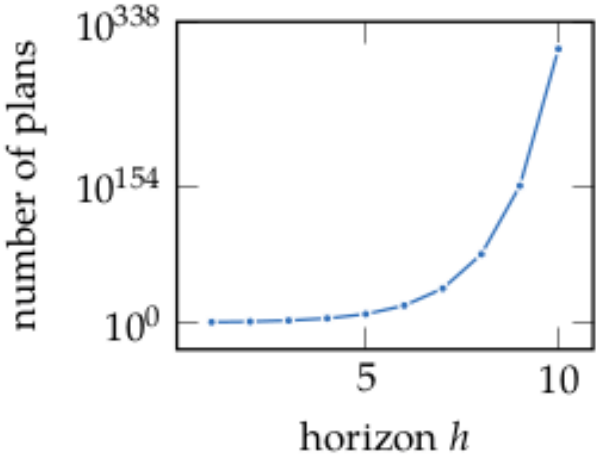
1 Step:

2 Step:



$$|A|^{\frac{(|O|^h-1)}{(|O|-1)}}$$

27 two step plans!



Alpha Vectors for Conditional Plans

For 1-step: $U^{\pi}(s) = R(s, \pi())$

$$U^{\pi}(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_{o} O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

POMDP Value Functions

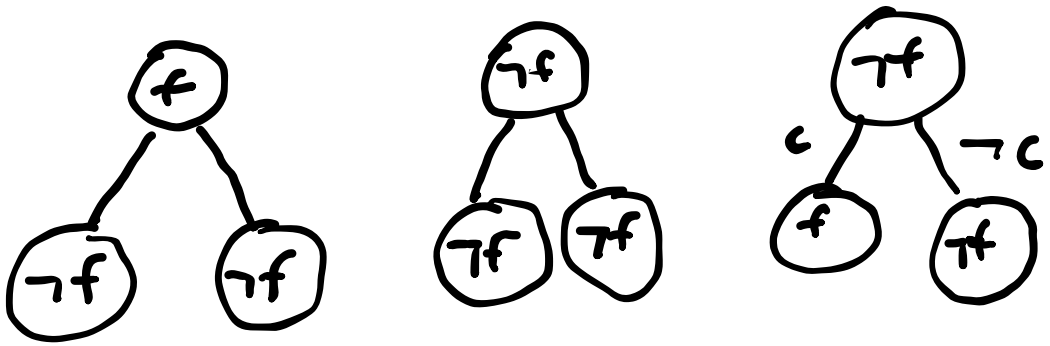
$$V^*(b) = \max_{\alpha \in \Gamma} \alpha^\top b$$

Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\} \qquad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \qquad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \qquad T(\neg h \mid \cdot, f) = 1.0$$



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$$Z(c \mid \cdot, h) = 0.8)$$

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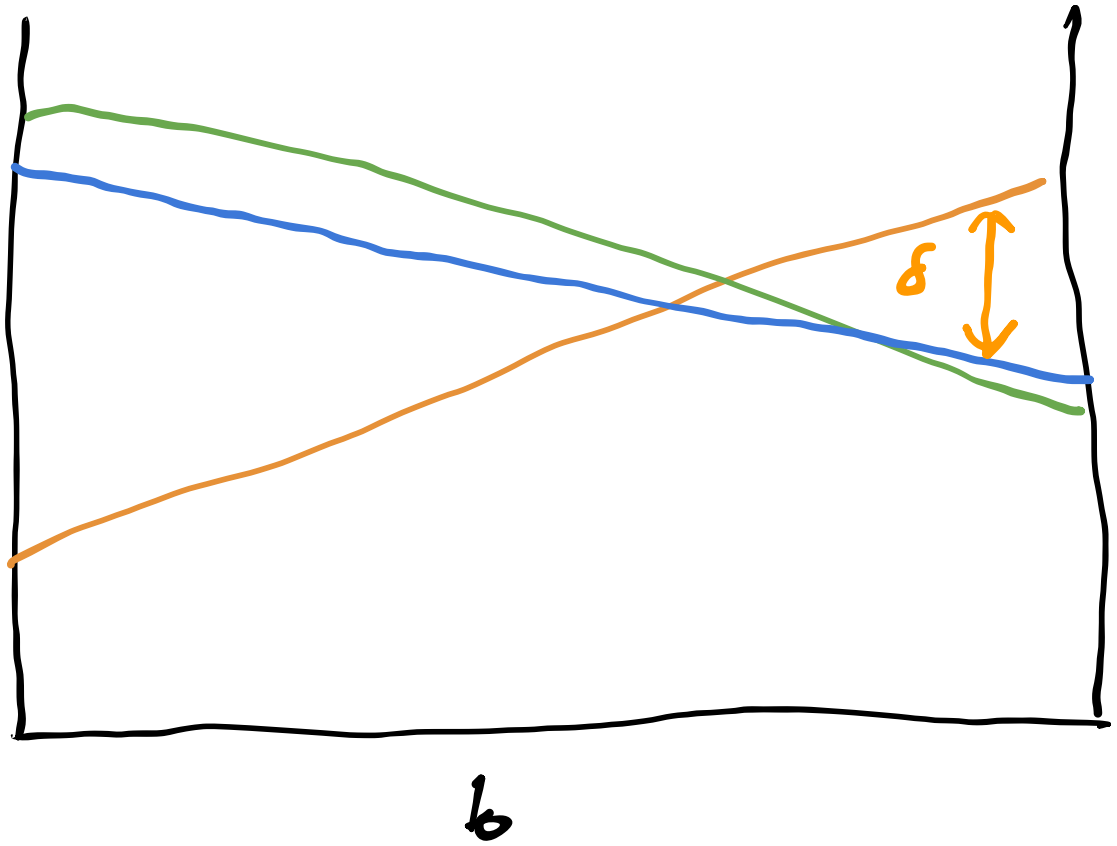
$$\gamma = 0.9$$

$$U^{\pi}(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_{o} O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

α-Vector Pruning



$$\begin{aligned} \underset{\delta, b}{\text{maximize}} \quad & \delta \\ \text{subject to} \quad & b \geq 0 \\ & \mathbf{1}^\top b = 1 \\ & \alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma \end{aligned}$$

"Linear Program": easy optimization problem

If $\delta^* > 0$, α is not dominated; b solution sometimes called "witness".

POMDP Value Iteration (horizon d)

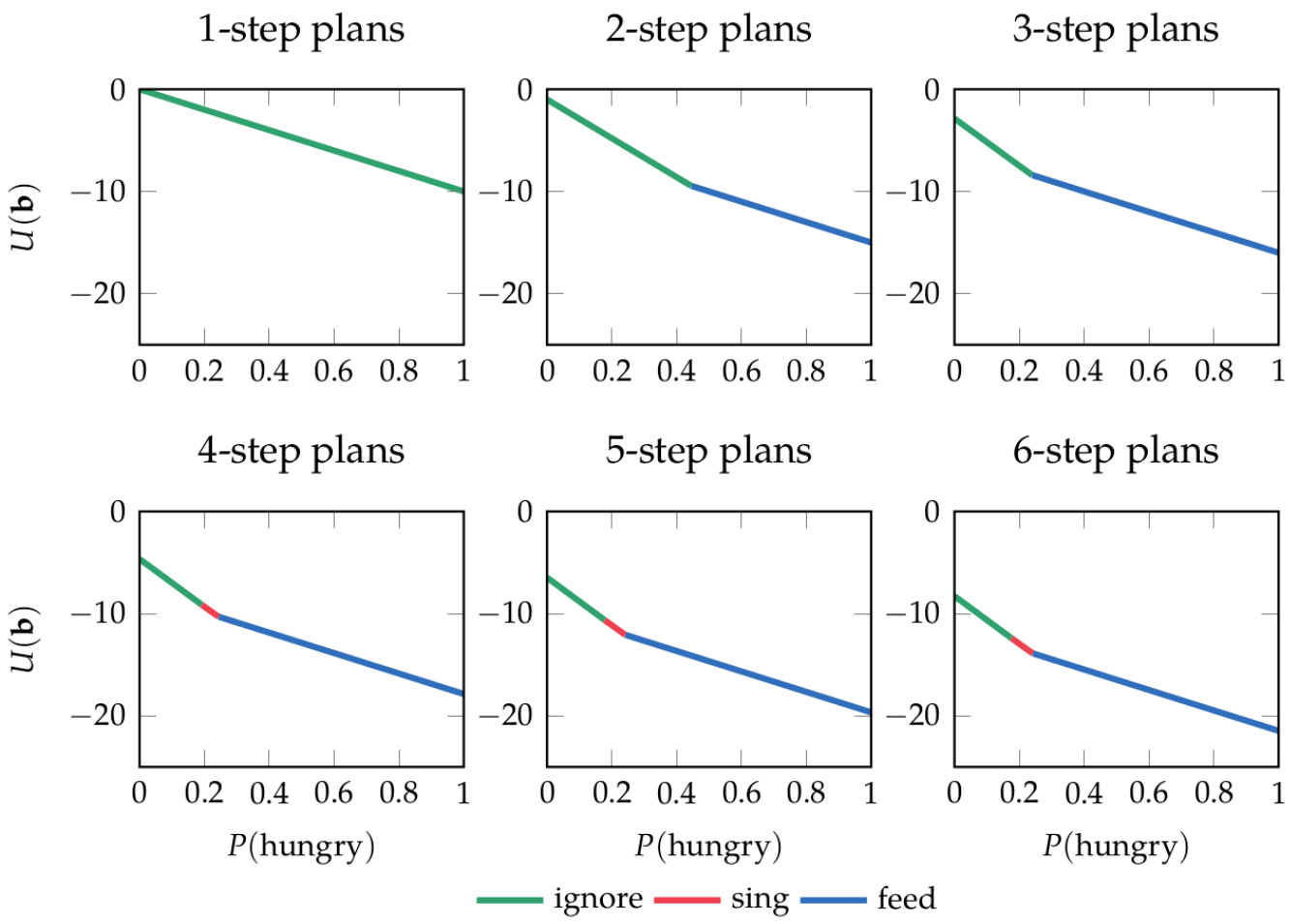
$\Gamma^0 \leftarrow \emptyset$

for $n \in 1 \ldots d$

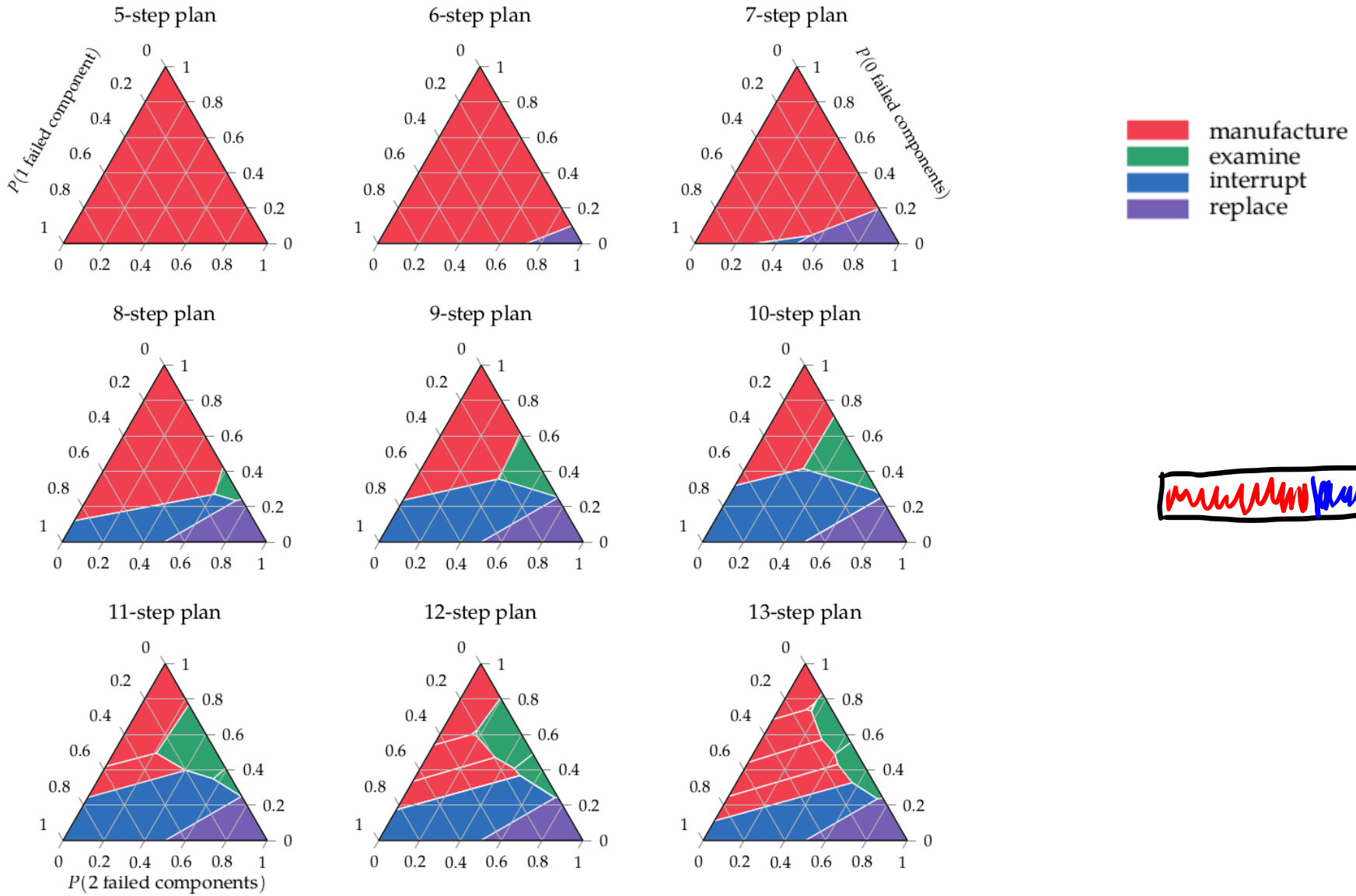
Construct Γ^n by expanding with Γ^{n-1}

Prune Γ^n

Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α-vectors
- Each α vector corresponds to a conditional plan